

# PakLaw AI

Pakistan AI Policy & Law Assistant — Short PRD

## 1. Project Overview

PakLaw AI is a Pakistan-focused AI legal information assistant. Users ask questions in natural language, and the system retrieves relevant information from pre-indexed Pakistani laws and policies using RAG, then generates a simple answer with sources.

## 2. Problem

- Pakistani laws are often lengthy and difficult to understand.
- Finding the correct section/article manually can take time.
- Users need a simple way to search legal information.

## 3. Proposed Solution

Create a web-based chatbot containing a curated collection of Pakistani legal documents. Documents are processed once and stored as searchable embeddings. At query time, only relevant legal passages are retrieved and provided to the LLM.

## 4. Core Workflow

| One-Time Data Preparation | User Query            |
|---------------------------|-----------------------|
| PDF Laws                  | Question              |
| Text Extraction           | Query Embedding       |
| Cleaning + Chunking       | Semantic Search       |
| Embeddings + Metadata     | Relevant Legal Chunks |
| Store in Database         | LLM / RAG             |
|                           | Answer + Sources      |

## 5. Initial Knowledge Base

- Constitution of Pakistan, 1973
- Pakistan Penal Code, 1860
- Code of Criminal Procedure, 1898
- Code of Civil Procedure, 1908
- Qanun-e-Shahadat Order, 1984
- Prevention of Electronic Crimes Act, 2016
- Police Order, 2002

## 6. Main Features

- Chat-based legal question answering
- RAG-based retrieval from Pakistani legal documents
- Section/article/page references where available
- Source display for answer verification
- English + optional Urdu/Roman Urdu questions

- Follow-up questions with basic conversation context
- Safe response when information is not available

## 7. Suggested Technology

- Frontend: HTML, Bootstrap, JavaScript/jQuery
- Backend: PHP
- LLM: Ollama + Gemma (local)
- Embeddings: Local embedding model
- Database: MySQL / vector-capable storage
- RAG: PDF extraction → chunking → embeddings → similarity search → LLM

## 8. MVP Scope

- 5–10 authoritative Pakistani legal documents
- One-time ingestion/indexing process
- Chat interface
- Semantic retrieval
- RAG answer generation
- Sources and legal references
- Basic error and no-result handling
- Legal information disclaimer

## 9. Development Steps

1. Finalize Pakistani legal documents and sources.
2. Build PDF text extraction.
3. Clean and structure legal text.
4. Chunk by section/article where possible.
5. Generate and store embeddings with metadata.
6. Implement similarity search.
7. Connect retrieved context to Gemma/LLM.
8. Build Bootstrap chat UI and PHP API.
9. Display answer + sources.
10. Test retrieval accuracy, hallucinations and response time.
11. Prepare demo and presentation.

## 10. Example Demo Questions

- Section 144 kya hai?
- Police mujhe warrant ke baghair kab arrest kar sakti hai?
- What are fundamental rights under the Constitution?
- Pakistan mein cybercrime ke liye konsa law apply hota hai?

## 11. Safety / Disclaimer

PakLaw AI provides legal information for educational and informational purposes only. It does not replace professional legal advice. The system should not invent laws, sections, articles or punishments and should state

when sufficient information is not available.

**Hackathon Goal:** Demonstrate a working Pakistan-focused RAG application that answers legal questions from pre-indexed Pakistani law documents and shows the supporting sources.